

Elijah E. Mullens
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Education

Ph.D. Astronomy, <i>Cornell University</i>	August 2027 (estimated)
Advisor: Prof. Nikole Lewis (nikole.lewis@cornell.edu)	
Dissertation Title: “Cloudy With a Chance of Silicates: An Exploration of Exoplanet Aerosols and Weather”	
M.S. Astronomy, <i>Cornell University</i>	November 2024
B.S. Mathematics, Physics, and Astrophysics, Cum Laude, <i>University of Florida</i>	May 2021
Minor in East Asian Languages and Literatures - Japanese	
A.S. Tallahassee Community College / Florida State University Associate of Arts	May 2017

Honors and Awards

Cranson W. and Edna B. Shelly Excellence in Research In Astronomy Award	2026
Cranson W. and Edna B. Shelly Outstanding Teaching Award, <i>Cornell Department of Astronomy</i>	2025
NASA Summer Internship Program	2025
NSF Graduate Research Fellowship	2023-2027
Graduate School Dean’s Excellence Fellowship, <i>Cornell University</i>	2022-2023
JASSO Scholarship, <i>Nagoya University</i>	2019-2020
Florida Bright Futures Scholarship - Academic Scholars Award	2017-2021
Presidential Scholarship, <i>University of Florida</i>	2017-2021
Dean’s List, President’s Honor Roll, <i>University of Florida</i>	2017-2021

Relevant Experience

NASA Goddard Space Flight Center	Summer 2025
Intern (PI: Dr. Sarah Moran)	
Project: “Experimental and Modeling Investigation of Exoplanet Cloud Properties”	
Cornell University, Department of Astronomy	2024-2025
Teaching Assistant/Head Teaching Assistant	
ASTRO 1195 (Observational Astronomy)	
Smaller class (30 students). Held night labs, graded homework, and gave guest lectures.	
ASTRO 1102 (Our Solar System)	
Large class (130 students). Head-TA with multiple duties such as: attending lecture, holding hours, developing material for discussion section, organizing TA duties	
Space Telescope Science Institute (STScI)	Summer 2022
Space Astronomy Summer Program Intern (PI: Dr. Catherine Zucker)	
Project: “Unveiling the Nature of Diffuse Interstellar Envelopes Around Dense Star-forming Clouds”	

Developed a pipeline to compare interstellar cloud simulations to observations, gave a presentation at the 2022 SASP Symposium, will give a poster presentation at AAS 241 and Protostars and Planets VII. Resulting paper accepted for publication in ApJ.

Challenger Learning Center of Tallahassee 2021-2022

Planetarium Instructor

Created planetarium shows using Digistar3, ran planetarium shows, prepared and presented live the monthly ‘Monthly Skies Over Tallahassee’ and other educational pre-shows (‘Monthly Astronomy Learning Topic’, ‘Space News’, ‘Monthly Solar System Object’).

Meitetsu Work Travels Inc. - Nagoya University 2019-2020

English Camp Tutor

Led and organized group activities for over 1,000 children. Gave presentation on home country and taught basic mathematics.

Nagoya University, Department of Physics 2019-2020

Visiting Undergraduate Researcher

Further topics in Statistical Physics with Professor John Wojdylo, covering microcanonical and canonical formalism, quantum fluids, and mean field theory.

Nagoya University, Department of Earth and Environmental Sciences 2019-2020

Visiting Undergraduate Researcher

Further topics in Earth and Planetary Sciences with Professor Marc Humblet, research paper written on TESS and planetary detection methods.

University of Florida, Department of Astronomy Fall 2020

Undergraduate Researcher

Contributed to Professor Charles Telesco’s IMPS (Integrated Miniature Polarimeter and Spectrograph) project. Achieved an understanding of the instrument while learning further topics on polarimetry and astrobiology.

University of Florida, Department of Mathematics Fall 2019

Undergraduate Researcher

Differential Geometry project with Professor Luca Di Cerbo covering Do Carmo’s Differential Geometry of Curves and Surfaces, with a presentation at the end.

Certifications

Florida State University

Global Partners Certificate (2021)

Equity, Diversity, and Inclusion Certificate (2021)

Coursera

The University of Sydney: Data-driven Astronomy (2021)

NUPACE Program

Nagoya University: Certificate of Completion (2020)

Approved Grants & Observing Proposals

JWST, Cycle 5, 6.8 hours (Co-I, PI Boehm, JWST-GO-12237) 2026-2029

“A Definitive Transmission Survey of the Atmosphere of 55 Cnc e”

JWST, Cycle 5, 9.8 hours (Co-I, PI Lothringer, JWST-GO-10290) 2026-2029

“Mistaken Identity? Resolving Ultra-hot Jupiter NUV Absorption to Measure Refractory-to-Volatile

Ratio”	
JWST, Cycle 4, 13.7 hours (Co-I, PI MacDonald, JWST-GO-09033) “By the Ashes of Stars: A Chemical Census of a White Dwarf Planet”	2025-2028
JWST, Cycle 4, 176.4 hours (Co-I, PI Faherty, JWST-GO-08441) “The Cold Worlds Spectral Library”	2025-2028
JWST, Cycle 4, 37.7 hours (PI, JWST-GO-08309) “It’s (poly)Morphin’ Time! Solving the quartz quandry of WASP-17b”	2025-2028
JWST, Cycle 4, 46.1 hours (Co-I, PI Moran, JWST-GO-07849) “Meeting their Match: Measuring Cloudy Hot Jupiter Counterparts to Understand the Silicate Cloud Regime with JWST/MIRI”	2025-2028
JWST, Cycle 4, 31.7 hours (PI, JWST-GO-07686) “Brown Dwarf Broiler: Probing Chemical Quenching and Heat Redistribution in a Highly-Eccentric Brown Dwarf”	2025-2028
NSF Graduate Research Fellowship (\$159,000, Advisor: Prof. Nikole Lewis) “Chemical Timescales in the Atmospheres of Highly Eccentric Sub-Neptunes”	2023-2027
Other Worlds Laboratory (OWL) (\$2000) In support of participation in OWL summer school	Summer 2025
Cornell Graduate School Research Travel Grant (\$2000) In support of internship at NASA Goddard Space Flight Center	Summer 2025
Cornell Graduate School Conference Grant (\$700) In support of presentation at 243rd American Astronomy Society Meeting	Winter 2024
Cornell Graduate School Conference Grant (\$700) In support of presentation at Protostars & Planets VII	Spring 2023

Presentations (Talks & Posters)

AOGS 2026, Accepted Talk “The Quartz Quandry of WASP-17b: Extracting Maximal Information from Silicate Cloud Features in Exoplanet Spectra Through Various Modeling and Laboratory Techniques”	August 2026
BOWIE+ Collaboration, Invited Speaker “Head in the (Silicate) Clouds: Silicate Sundogs and Grain Alignment”	February 2026
Modern Language Association (MLA), Accepted Talk “An idea you’ll have to give birth to’: An exploration of astronomy and alternative relationality in Planet-Size X-Men”	January 2026
American Geophysical Union (AGU), Accepted Talk “HEL-Fire and Brimstone: High Temperature Optical Properties of Quartz and Tridymite for Substellar Cloud Studies”	December 2025
American Astronomical Society (AAS246), Accepted iPoster “Silicate Sundogs and Sandy Thunderstorms: Probing the Effects of Grain Directionality and Chemistry in Exoplanet Observations”	June 2025
American Astronomical Society (AAS246), LabAstro Workshop Lightning Talk	June 2025

“The Quartz Quandary of WASP-17b”	
Emerging Researchers in Exoplanet Science (ERESIX), Accepted Talk	July 2024
“A New Atmospheric Retrieval Paradigm for Exoplanet Aerosols”	
Outer Planets Assessment Group (OPAG), Lightning Talk+Poster	June 2024
“Life After Death: Europa in the Habitable Zone of a Red Sun and White Dwarf”	
Astrobiology Graduate Conference (AbGradCon), Accepted Poster	June 2024
“Life After Death: Europa in the Habitable Zone of a Red Sun and White Dwarf”	
American Astronomical Society (AAS243), Accepted iPoster	January 2024
“A New Atmospheric Retrieval Paradigm for Exoplanet Aerosols”	
American Astronomical Society (AAS241), Accepted iPoster	January 2023
“Unveiling the Nature of Diffuse Interstellar Envelopes Around Dense Star-forming Clouds”	
Protostars and Planets 7, Accepted Poster	April 2023
“Unveiling the Nature of Diffuse Interstellar Envelopes Around Dense Star-forming Clouds”	

Professional Service & Associations

Kyoto University Retrieval Workshop, <i>Organizer</i>	2026
Undergraduate Mentor (Sophia Gammel)	2026-present
Referee for Elsevier	2026-present
Graduate Mentor (Isabela Huckabee)	2024-present
Undergraduate Mentor (Ian Branigan)	2024-present
AbGradCon 2024, <i>SOC Member</i>	2023-2024
Florida State University (FSU) Astronomy Club, <i>Founder</i>	2022
Astronomers for Planet Earth, <i>Member</i>	2021-2022
oSTEM Global Chapter, <i>Member</i>	2021-2022
Circle at Nagoya University “Ametama”, <i>Member</i>	2019-2020
Nagoya University Language Exchange Club, <i>Member</i>	2019-2020
Nagoya University Foreign Students Club, <i>Member</i>	2019-2020
oSTEM University of Florida, <i>Member</i>	2018-2019
University of Florida Japanese Language and Culture Club, <i>Member</i>	2017-2018
University of Florida Model UN (GatorMUN), <i>Member</i>	2017-2018

Public Outreach

Cornell University Department of Astronomy Outreach, <i>Volunteer</i>	2023-present
Graduate Student School Outreach Program (GRASHOPR), <i>Volunteer</i>	2023-2024
Sea-to-See Marine Biology Outreach Program, <i>Volunteer</i>	Summer 2022
Challenger Learning Center of Tallahassee, <i>Volunteer</i>	2021-2022
Tallahassee Astronomical Society, <i>Volunteer</i>	2021-2022
Tallahassee Scientific Society, <i>Event Organizer</i>	2021-2022

Selected Media

“Moment of Science: “Sun dogs” are possible on other planets, new study suggests”, *ABC News*, 8/28/2025
 “Sun dogs, other celestial effects could appear in alien skies”, *Cornell Chronicle*, 7/30/2025
 “Cornell astronomers win time on James Webb Space Telescope”, *Cornell Chronicle*, 6/17/2025
 “When the sun dies, could life survive on the Jupiter ocean moon Europa?”, *Space.com*, 6/03/2025
 “Will Europa Become a New Habitable World When the Sun Becomes a Red Giant?”, *Universe Today*, 5/27/2025

Publications

–6 first author manuscripts

Submitted/In preparation

E. Mullens et al. (2026) “Rocky Planet Picture Show: The Rocky Planet Picture Show: Implementation of Surface Reflection and Emission in POSEIDON with Application to and Interpretation of JWST Data”. *Submitted to PSJ*

E. Mullens et al. (2026) “Temperature-Dependent Light Transmission Properties of Silica Aerosols”, *Submitted to Nature Astronomy*

E. Mullens et al. (2026) “HEL-Fire and Brimstone: Application of Temperature Dependent Particulate Crystalline Quartz and Amorphous Silica Optical Properties to Substellar Cloud Studies”, *In Prep*

E. Mullens et al. (2026) “Tricky Tridymite: Temperature-Dependent Optical Properties and Transmittance of Particulate Synthetic and Natural Tridymite with an Application to Substellar Cloud Studies”, *In Prep*

Refereed Journal Articles

- [17] A. S. Zelakiewicz, **Mullens, Elijah**, L. Kaltenegger, and D. Savransky. Identifying Surface Degeneracies in Single-Visit Reflected Light Observations of Modern Earth using the Habitable Worlds Observatory. *arXiv e-prints*, page arXiv:2603.25694, Mar. 2026, 2603.25694.
- [16] R. J. MacDonald, C. E. O’Connor, V. A. Boehm, E. M. May, D. K. Sing, **Mullens, Elijah**, L. C. Mayorga, T. O. Foote, S. Blouin, L. A. Pearce, N. K. Lewis, J. Valenti, N. E. Batalha, M. Lally, J. D. Lothringer, M. S. Marley, I. Mishra, and S. E. Mullally. Aerosols and hydrocarbons in the atmosphere of a white dwarf planet. *Nature*, 655(8121):76–80, July 2026.
- [15] K. L. Chubb, D. Grant, H. R. Wakeford, S. E. Moran, N. E. Batalha, A. Egan, C. Fairman, D. Powell, K. B. Stevenson, L. Alderson, P. Gao, T. Kataria, N. K. Lewis, R. J. MacDonald, M. Marley, **Mullens, Elijah**, D. K. Sing, and J. A. Valenti. Magnesium Silicate Clouds in the Atmosphere of HD 209458b from a Rule-Based Tree-Structured Data Reduction. *arXiv e-prints*, page arXiv:2606.00177, May 2026, 2606.00177.
- [14] **Mullens, Elijah** and N. K. Lewis. Silicate Sundogs: Probing the Effects of Grain Directionality in Exoplanet Observations. *ApJ*, 988(2):L43, Aug. 2025, 2508.01839.
- [13] **Mullens, Elijah**, B. Schmidt, L. Kaltenegger, and N. K. Lewis. Life after death: Europa in the evolving habitable zone of a Red Sun. *MNRAS*, 540(1):1329–1344, June 2025, 2505.15495.
- [12] K. S. Sotzen, J. Lustig-Yaeger, K. B. Stevenson, S.-M. Tsai, R. C. Challener, J. Goyal, N. K. Lewis, D. R. Louie, L. C. Mayorga, D. Valentine, H. R. Wakeford, L. Alderson, N. H. Allen, T. J. Fauchez, A. Glidden, A. Gressier, S. M. Hörst, J. Huang, Z. Lin, A. M. Mandell, **Mullens, Elijah**, S. Peacock,

- E. W. Schwieterman, J. A. Valenti, C. M. Mountain, M. Perrin, R. P. van der Marel, Consortium On Habitability, and Atmospheres Of M-Dwarf Planets (Champs). Non-detection of Nightside Emission from HAT-P-26b due to Stellar Contamination. *Research Notes of the American Astronomical Society*, 9(12):352, Dec. 2025.
- [11] W. W. Meynardie, M. R. Meyer, R. J. MacDonald, P. Calissendorff, **Mullens, Elijah**, G. M. Zarazua, A. Roy, H. Ganta, E. C. Gonzales, A. Adams, N. Lewis, Y. Hong, and J. Lunine. Ross 458 C: Gas Giant or Brown Dwarf? *ApJ*, 994(2):237, Dec. 2025, 2509.22803.
- [10] J. Lustig-Yaeger, K. S. Sotzen, K. B. Stevenson, S.-M. Tsai, R. C. Challener, J. Goyal, N. K. Lewis, D. R. Louie, L. C. Mayorga, D. Valentine, H. R. Wakeford, L. Alderson, N. H. Allen, T. J. Fauchez, A. Glidden, A. Gressier, S. M. Hörst, J. Huang, Z. Lin, A. M. Mandell, **Mullens, Elijah**, S. Peacock, E. W. Schwieterman, J. A. Valenti, C. M. Mountain, M. Perrin, and R. P. van der Marel. JWST-TST DREAMS: The Nightside Emission and Chemistry of WASP-17b. *ApJ*, 994(1):L4, Nov. 2025, 2510.06169.
- [9] D. R. Louie, **Mullens, Elijah**, L. Alderson, A. Glidden, N. K. Lewis, H. R. Wakeford, N. E. Batalha, K. D. Colón, A. Gressier, D. Long, M. Radica, N. Espinoza, J. Goyal, R. J. MacDonald, E. M. May, S. Seager, K. B. Stevenson, J. A. Valenti, N. H. Allen, C. I. Cañas, R. C. Challener, D. Grant, J. Huang, Z. Lin, D. Valentine, M. Clampin, M. Perrin, L. Pueyo, R. P. van der Marel, and C. M. Mountain. JWST-TST DREAMS: A Precise Water Abundance for Hot Jupiter WASP-17b from the NIRISS SOSS Transmission Spectrum. *AJ*, 169(2):86, Feb. 2025, 2412.03675.
- [8] A. Gressier, R. J. MacDonald, N. Espinoza, H. R. Wakeford, N. K. Lewis, J. Goyal, D. R. Louie, M. Radica, N. E. Batalha, D. Long, E. M. May, **Mullens, Elijah**, S. Seager, K. B. Stevenson, J. A. Valenti, L. Alderson, N. H. Allen, C. I. Cañas, R. C. Challener, K. Colón, A. Glidden, D. Grant, J. Huang, Z. Lin, D. Valentine, C. M. Mountain, L. Pueyo, M. D. Perrin, and R. P. van der Marel. JWST-TST DREAMS: A Supersolar Metallicity in WASP-17 b’s Dayside Atmosphere from NIRISS SOSS Eclipse Spectroscopy. *AJ*, 169(2):57, Feb. 2025, 2410.08149.
- [7] A. Gressier, N. E. Batalha, N. Wogan, L. Alderson, D. Doud, N. Espinoza, R. J. MacDonald, H. R. Wakeford, J. A. Valenti, N. K. Lewis, S. Seager, K. B. Stevenson, N. H. Allen, C. I. Cañas, R. C. Challener, A. Glidden, J. Huang, Z. Lin, D. R. Louie, C. Maguire, **Mullens, Elijah**, K. Sotzen, D. Valentine, M. Clampin, L. Pueyo, R. P. van der Marel, and C. M. Mountain. JWST-TST DREAMS: Sulfur Dioxide in the Atmosphere of the Neptune-mass Planet HAT-P-26 b from NIRSpec G395H Transmission Spectroscopy. *AJ*, 170(5):292, Nov. 2025, 2509.16082.
- [6] A. Glidden, S. Ranjan, S. Seager, N. Espinoza, R. J. MacDonald, N. H. Allen, C. I. Cañas, D. Grant, A. Gressier, K. B. Stevenson, N. E. Batalha, N. K. Lewis, D. Long, H. R. Wakeford, L. Alderson, R. C. Challener, K. Colón, J. Huang, Z. Lin, D. R. Louie, **Mullens, Elijah**, K. Sotzen, J. A. Valenti, D. Valentine, M. Clampin, C. M. Mountain, M. Perrin, and R. P. van der Marel. JWST-TST DREAMS: Secondary Atmosphere Constraints for the Habitable Zone Planet TRAPPIST-1 e. *ApJ*, 990(2):L53, Sept. 2025, 2509.05407.
- [5] N. Espinoza, N. H. Allen, A. Glidden, N. K. Lewis, S. Seager, C. I. Cañas, D. Grant, A. Gressier, S. Courreges, K. B. Stevenson, S. Ranjan, K. Colón, B. M. Morris, R. J. MacDonald, D. Long, H. R. Wakeford, J. A. Valenti, L. Alderson, N. E. Batalha, R. C. Challener, J. Huang, Z. Lin, D. R. Louie, **Mullens, Elijah**, D. Valentine, C. M. Mountain, L. Pueyo, M. D. Perrin, A. Bellini, J. Kammerer, M. Libralato, I. Rebollido, E. Rickman, S. T. Sohn, and R. P. van der Marel. JWST-TST DREAMS: NIRSpec/PRISM Transmission Spectroscopy of the Habitable Zone Planet TRAPPIST-1 e. *ApJ*, 990(2):L52, Sept. 2025, 2509.05414.

- [4] D. Valentine, H. R. Wakeford, R. C. Challener, N. E. Batalha, N. K. Lewis, D. Grant, **Mullens, Elijah**, L. Alderson, J. Goyal, R. J. MacDonald, E. M. May, S. Seager, K. B. Stevenson, J. A. Valenti, N. H. Allen, N. Espinoza, A. Glidden, A. Gressier, J. Huang, Z. Lin, D. Long, D. R. Louie, M. Clampin, M. Perrin, R. P. van der Marel, and C. M. Mountain. JWST-TST DREAMS: Nonuniform Dayside Emission for WASP-17b from MIRI/LRS. *AJ*, 168(3):123, Sept. 2024, 2410.08148.
- [3] **Mullens, Elijah**, C. Zucker, C. E. Murray, and R. Smith. Characterizing the 3D Structure of Molecular Cloud Envelopes in the Cloud Factory Simulations. *ApJ*, 966(1):127, May 2024, 2403.03112.
- [2] **Mullens, Elijah**, N. K. Lewis, and R. J. MacDonald. Implementation of Aerosol Mie Scattering in POSEIDON with Application to the Hot Jupiter HD 189733 b’s Transmission, Emission, and Reflected Light Spectrum. *ApJ*, 977(1):105, Dec. 2024, 2410.19253.
- [1] D. Grant, N. K. Lewis, H. R. Wakeford, N. E. Batalha, A. Glidden, J. Goyal, **Mullens, Elijah**, R. J. MacDonald, E. M. May, S. Seager, K. B. Stevenson, J. A. Valenti, C. Visscher, L. Alderson, N. H. Allen, C. I. Cañas, K. Colón, M. Clampin, N. Espinoza, A. Gressier, J. Huang, Z. Lin, D. Long, D. R. Louie, M. Peña-Guerrero, S. Ranjan, K. S. Sotzen, D. Valentine, J. Anderson, W. O. Balmer, A. Bellini, K. K. W. Hoch, J. Kammerer, M. Libralato, C. M. Mountain, M. D. Perrin, L. Pueyo, E. Rickman, I. Rebollido, S. T. Sohn, R. P. van der Marel, and L. L. Watkins. JWST-TST DREAMS: Quartz Clouds in the Atmosphere of WASP-17b. *ApJ*, 956(2):L32, Oct. 2023, 2310.08637.